

***REPORT
ON
PROJECT***

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Clicks to Conversions: Google Ads Sales Analysis through Python-Powered Insights and Power BI Visualization

Tools Used:

- Python
- Power BI

Submission Date: 15-June -2026

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Raw Dataset Link: [Google Ads Sales Analysis Dataset – Raw Data](#)

Project Link: [Google Ads Sales Analysis Python Code](#)

Clicks to Conversions: Google Ads Sales Analysis through Python-Powered Insights and Power BI Visualization

1. Project Overview

Developed an interactive Google Ads analytics dashboard using Python for data preprocessing and Power BI for visualization. The project analyzes advertising campaign data for data analytics courses and services, addressing challenges such as typos, formatting issues, missing values, and inconsistencies. Through data cleaning, transformation, and visualization, the dashboard uncovers patterns in clicks, impressions, conversions, and sales revenue, helping identify high-ROI keywords, device performance, and seasonal trends. This enables marketers to optimize targeting strategies, improve cost efficiency, and support data-driven decision-making for future campaigns.

Project Objective

- To analyze Google Ads campaign data and uncover meaningful patterns and trends.
- To clean and preprocess the dataset for accurate and reliable insights.
- To design impactful Power BI visualizations that clearly communicate ad performance.
- To evaluate key KPIs such as CTR, CPC, ROAS, and ROI for measuring effectiveness.
- To compare performance across Location, Device, and Keyword for better targeting.
- To document the workflow and highlight limitations for transparency and reproducibility.

2. Data Sources

Source Description and Timeline: [Google Ads Sales Analysis Dataset](#)

Domain: Sales and E-commerce

Year / Timeline: 2024

3. Problem Statement

Google Ads campaign data is often raw, inconsistent, and difficult to interpret due to issues like typos, missing values, and formatting errors. This project focuses on cleaning and transforming the dataset using Python, followed by creating interactive dashboards in Power BI to visualize campaign performance. The analysis aims to uncover trends in clicks, conversions, and ROI across keywords, devices, and locations. Ultimately, the project provides actionable insights to optimize targeting strategies, improve cost efficiency, and support data-driven marketing decisions.

4. Attribute (Column /Features) Details

Attribute Name	Data Type	Description
Ad_ID	Object (String)	Unique ID of the ad campaign
Campaign_Name	Object (String)	Name of the campaign (with typos and variations)
Clicks	Float64 (Numeric)	Number of clicks received
Impressions	Float64 (Numeric)	Number of ad impressions
Cost	Object (String)	Total cost of the ad (in ₹ or \$ format with missing values)
Leads	Float64 (Numeric)	Number of leads generated
Conversions	Float64 (Numeric)	Number of actual conversions (signups, sales, etc.)
Conversion Rate	Float64 (Numeric)	Calculated conversion rate (Conversions ÷ Clicks)
Sale_Amount	Object (String)	Revenue generated from the conversions

Ad_Date	Object (String/Date)	Date of the ad activity (in inconsistent formats like YYYY/MM/DD, DD-MM-YY)
Location	Object (String)	City where the ad was served (includes spelling/case variations)
Device	Object (String)	Device type (Mobile, Desktop, Tablet with mixed casing)
Keyword	Object (String)	Keyword that triggered the ad (with typos)

5. Tools & Technologies

- **Python:** Data cleaning, preprocessing, transformation, and visualization.
- **Power BI:** Visualization, and interactive dashboard creation.

6. Data Pre-Processing (Python)

Tasks Performed:

Data Cleaning & Transformation: Removed duplicates, handled missing values, standardized formats, Splitting, Find and replace.

Correcting Keyword Typos:

In the Campaign_Name and Keyword fields, multiple misspellings such as “DataAnalyticsCourse,” “Data Anlytics Corse,” “analitics,” and “anaytics” were corrected and standardized. All values were then capitalized to ensure consistency and improve the reliability of subsequent analysis.

```

# Fix Typos in Keyword column and Capitalize each word
ad_df['Keyword'] = ad_df['Keyword'].str.replace(r"(analitics|anaytics)", "Analytics", regex=True)
ad_df['Keyword'] = ad_df['Keyword'].str.title()
ad_df['Keyword'].unique()

array(['Learn Data Analytics', 'Data Analytics Course',
      'Data Analytics Online', 'Data Analytics Training',
      'Online Data Analytic', 'Analytics For Data'], dtype=object)

# Fix Typos in Campaign Name
ad_df["Campaign_Name"] = ad_df["Campaign_Name"].str.replace(
    r"(DataAnalyticsCourse|Data Anlytics Corse|Data Analytcis Course|Data Analytics Corse)",
    "Data Analytics Course",
    regex=True
)
ad_df['Campaign_Name'].str.capitalize()
ad_df['Campaign_Name'].unique()

array(['Data Analytics Course'], dtype=object)

```

Standardizing Text Fields:

In the *Location* column, typographical errors such as “Hyderbad” and “Hydrebad” were corrected and standardized to “Hyderabad.” Similarly, entries in the *Device* column were capitalized to ensure uniformity across categories like Desktop, Mobile, and Tablet.

```

# Fix Inconsistency in Location Column - Capitalize and fixing Typos
ad_df['Location'] = ad_df['Location'].str.capitalize()
ad_df['Location'] = ad_df['Location'].str.replace(r"(Hyderbad|Hydrebad)", "Hyderabad", regex=True)
ad_df['Location'].unique()

array(['Hyderabad'], dtype=object)

# Fix Inconsistency in Device Column - Capitalize
ad_df['Device'] = ad_df['Device'].str.capitalize()
ad_df['Device'].unique()

array(['Desktop', 'Mobile', 'Tablet'], dtype=object)

```

Standardizing Date Formats:

In the *Ad_Date* column, inconsistent date entries were cleaned by stripping extra spaces and safely parsing them into a uniform format. This ensured all dates were standardized for reliable analysis. The code removes whitespace, defines a *safe_parse* function to handle errors, and applies it to convert text entries into proper date objects. As a result, all values in *Ad_Date* are stored consistently in YYYY-MM-DD format.

```
# Fixing Data Format
from dateutil import parser

# Clean and standardize text (removing white spaces)
ad_df['Ad_Date'] = ad_df['Ad_Date'].astype(str).str.strip()

# Define a safe_parse function
def safe_parse(date_str):
    try:
        # Try parsing each date, interpret the string as a date
        return parser.parse(date_str, dayfirst=True).date()
    except Exception:
        return pd.NaT

# Apply parsing function
ad_df['Ad_Date'] = ad_df['Ad_Date'].apply(safe_parse)

ad_df['Ad_Date']
```

	Ad_Date
0	2024-11-16
1	2024-11-20

Handling Missing Values and Outliers:

Numeric fields such as *Clicks*, *Impressions*, *Cost*, *Leads*, *Conversions*, and *Sale_Amount* were imputed using either the mean or median, depending on skewness and presence of outliers. This ensured that missing values were replaced appropriately without distorting the distribution. The function checks skewness and detects outliers using the IQR method. If the data is symmetric and free of outliers, missing values are filled with the mean; otherwise, the median is used. This balances accuracy with robustness in the imputation process.

```
# Code Function
def choose_imputation(series):
    # remove missing values
    data = series.dropna()
    # Calculate skewness to measures symmetry of the distribution
    skew = data.skew()
    # Detect outliers using IQR
    Q1 = data.quantile(0.25)
    Q3 = data.quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = data[(data < lower) | (data > upper)]
    # Deciding Mean or median
    if abs(skew) < 0.5 and len(outliers) == 0:
        method = "mean"
        value = data.mean()
    else:
        method = "median"
        value = data.median()

    return method, value

# Numeric columns except Conversion Rate(as it need seprate calculation)
numeric_cols = ['Clicks','Impressions','Cost','Leads','Conversions','Sale_Amount']

for col in numeric_cols:
    method, value = choose_imputation(ad_df[col])
    #print(f"{col}: using {method}, value = {value}")
    ad_df[col] = round(ad_df[col].fillna(value),2)
#Check for null
ad_df[['Clicks','Impressions','Cost','Leads','Conversions','Sale_Amount']].isnull().sum()
```

0

Clicks 0

Impressions 0

Calculated Columns & Python Metrics:

Implemented Python formulas to derive key advertising KPIs, including cost per acquisition (CPA), cost per click (CPC), cost per lead (CPL), revenue per conversion (RPC), sales ROI, click-through rate (CTR), conversion rate (CVR), lead conversion rate (LCR), and profit. These calculated fields provide deeper insights into campaign performance and efficiency.

```
# Calculate Cost per Acquisition (CPA)
ad_df['CPA'] = round(ad_df['Cost'] / ad_df['Conversions'],2)

# Calculate Cost per click (CPC)
ad_df['CPC'] = round(ad_df['Cost'] / ad_df['Clicks'],2)

# Calculate Cost per lead (CPL)
ad_df['CPL'] = round(ad_df['Cost'] / ad_df['Leads'],2)

# Calculate Revenue per conversion (RPC)
ad_df['RPC'] = round(ad_df['Sale_Amount'] / ad_df['Conversions'],2)

# Calculate Sales Efficiency(Return on Spend,Revenue Efficiency,Revenue-to-Cost Ratio)
ad_df['Sales_ROI'] = round(ad_df['Sale_Amount'] / ad_df['Cost'],2)

# Calculate CTR (Click-Through Rate)
ad_df['CTR'] = round(ad_df['Clicks'] / ad_df['Impressions'],2)

# Calculate CTR (%)
ad_df['CTR%'] = round((ad_df['Clicks'] / ad_df['Impressions']) * 100,2)

# Calculate Conversion Rate (CVR)
ad_df['CVR'] = round(ad_df['Conversions'] / ad_df['Clicks'],2)
# CVR (%)
ad_df['CVR%'] = round((ad_df['Conversions'] / ad_df['Clicks']) * 100,2)

# Calculate Lead Conversion Rate (LCR)
ad_df['LCR'] = round(ad_df['Conversions'] / ad_df['Leads'],2)
# Calculate Lead Conversion Rate Percent (LCR%)
ad_df['LCR%'] = round(ad_df['LCR'] * 100,2)

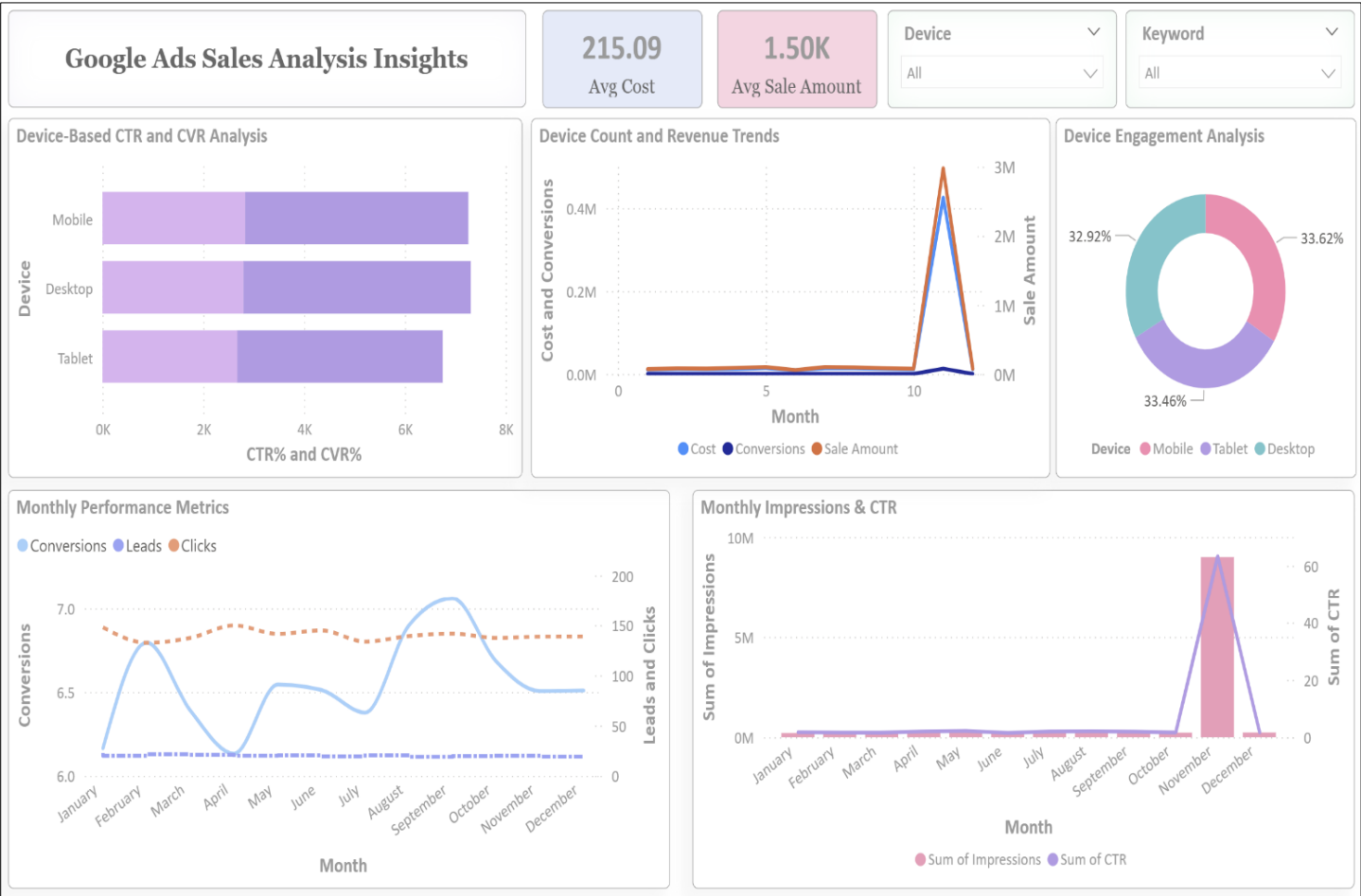
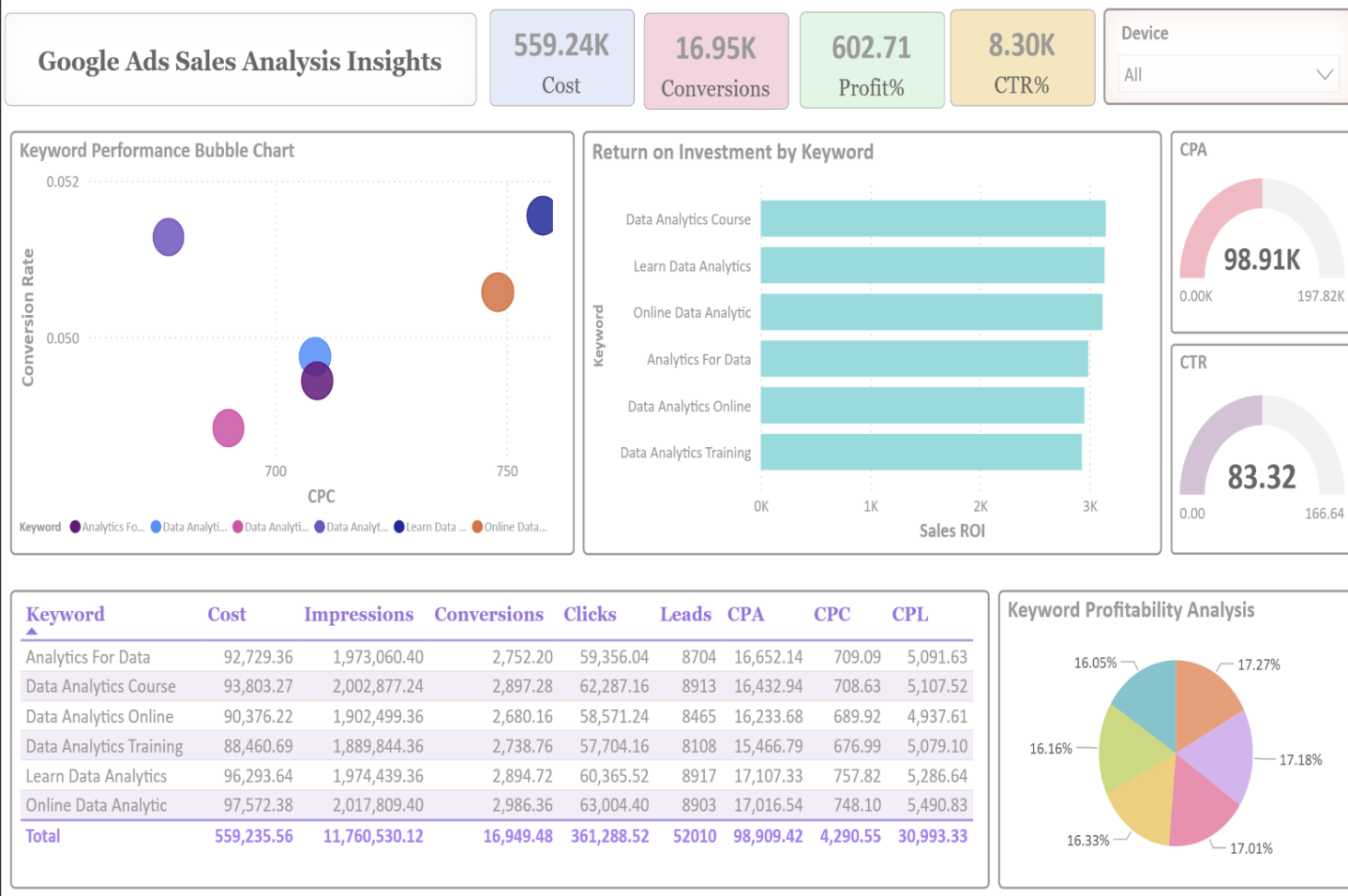
# Calculate Profit (Return on Investment)
ad_df['Profit'] = round((ad_df['Sale_Amount'] - ad_df['Cost']) / ad_df['Cost'],2)
# Calculate Profit%
ad_df['Profit%'] = round(((ad_df['Sale_Amount'] - ad_df['Cost']) / ad_df['Cost']) * 100,2)

ad_df[['CPA','CPC','CPL','RPC','Sales_ROI','CTR','CTR%','CVR','CVR%','LCR','LCR%','Profit','Profit%']]
```

7. Visualizations (Power BI) Dashboard Features

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, donut chart, funnel chart, column chart, scatter chart, tree mab, gauge and tables to communicate insights.

SCREENSHOT OF DASHBOARD



8. Insights & Conclusions

▪ Key Findings:

▪ Provide the analysis insights:

▪ Descriptive

- Summarizes the Google Ads campaign dataset to provide an overall view of performance
- Analyzes key metrics such as clicks, impressions, conversions, cost, and sales revenue using summary statistics (mean, median, maximum, minimum)
- Applies frequency counts to measure occurrences, for example the number of campaigns by Location or Device
- Uses simple visualizations such as bar charts, pie charts, and line graphs to present campaign performance and highlight trends
- Outcome: Provides a comprehensive snapshot of “what happened” across campaigns.

▪ Diagnosis

- Investigates the underlying reasons behind observed patterns in campaign performance
- Uses correlation analysis to explore relationships between variables, such as the link between advertising cost and conversions
- Applies comparative analysis to evaluate performance across dimensions like Location, Device, and Keyword
- Employs drill down dashboards in Power BI to isolate underperforming campaigns and uncover specific areas for improvement
- Outcome: Provides actionable insights into “why it happened”, guiding optimization strategies such as reallocating budget to high performing keywords or devices

▪ Predictive

- Uses historical Google Ads campaign data to forecast future performance trends.
- Applies regression models and time-series analysis to estimate expected clicks, conversions, and ROI.
- Identifies seasonal patterns and keyword behaviors that influence campaign

outcomes.

- Employs Power BI forecasting visuals to project future sales and conversion rates.
- Outcome: Provides insights into “what is likely to happen,” enabling proactive planning and budget allocation for upcoming campaign

- **Prescriptive**

- Builds on diagnostic and predictive insights to recommend optimal actions.
- Uses scenario modelling and optimization techniques to suggest budget reallocation across keywords, devices, and locations.
- Applies decision rules and ROI thresholds to prioritize high-performing campaigns.
- Employs Power BI “What-If” parameters to simulate different strategies and outcomes.
- Outcome: Provides actionable guidance on “what should be done,” helping marketers maximize ROI, reduce costs, and improve targeting efficiency.

9. Conclusions

This project successfully analyzed Google Ads campaign data using Python for preprocessing and Power BI for visualization to understand advertising performance, cost efficiency, and sales outcomes. The dashboard provided meaningful insights into:

- Keyword effectiveness
- Device performance
- Seasonal campaign trends
- Conversion efficiency
- Revenue generation

Through interactive visualizations and data analysis techniques, the project demonstrated how digital marketers can use business intelligence tools to optimize targeting strategies, improve ROI, and support better decision-making. The project also highlighted the importance of marketing analytics in identifying profitable areas, refining keyword prioritization, enhancing mobile optimization, and planning seasonal campaigns to maximize overall performance.

10. Future Enhancement

- **Predictive Modeling:** Apply machine learning techniques to forecast future clicks, conversions, and revenue using historical campaign data.
- **Real Time Monitoring:** Integrate dashboards with the Google Ads API for live performance tracking and automated KPI alerts.
- **Advanced Segmentation:** Refine targeting strategies by segmenting audiences based on demographics, behavior, and geography.
- **Optimization Algorithms:** Implement smart budget allocation and bid optimization models to maximize conversions while reducing costs.
- **Cross Channel Analytics:** Expand analysis to include social media, email, and SEO for a unified view of digital marketing performance.
- **Scalable Dashboards:** Enhance dashboards with drill down features, automated updates, and cloud integration for enterprise scalability.